

# MADISON EMSHOUSEN

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## PROFESSIONAL SUMMARY

Insider Threat Investigator and ML Engineer with 3+ years of experience conducting end-to-end security investigations, building machine learning pipelines, and automating forensic workflows at enterprise scale. Proven ability to correlate endpoint, identity, and data-loss signals across CrowdStrike, Microsoft Purview, and Defender to reconstruct attacker and insider activity. Adept at translating raw data into clear, evidenced findings for legal, HR, and security leadership.

## TECHNICAL SKILLS

**Languages:** Python, SQL, KQL, PowerShell, C++, Java, JavaScript, C#, R, Lua, HTML  
**ML / AI:** Machine Learning, Large Language Models (LLMs), Predictive Modeling, Anomaly Detection, NLP, MLOps  
**Data & Analytics:** ETL Pipelines, Data Engineering, Dashboard Development, Data Visualization, Matplotlib, Pandas  
**Databases:** SQL (advanced), RDBMS, Query Optimization, Indexing  
**Security & Investigations:** Insider Threat Investigations, Data Loss Prevention, Digital Forensics, Fraud Detection, Incident Response, OSINT, Chain of Custody, Threat Analysis  
**Security Tools:** CrowdStrike Falcon, Microsoft Purview, Microsoft Defender, Microsoft Entra ID, Intune, Power Automate  
**Tools & Platforms:** Jupyter, Git/GitHub, React.js, Tableau/BI tools, Agile

## EXPERIENCE

**Allstate** Aug 2026 – Present  
*Insider Threat Investigator · Remote*

- Investigate potential insider threats — including data exfiltration, policy violations, and access misuse — from first alert through documented findings, correlating endpoint telemetry from CrowdStrike Falcon and Microsoft Defender with Microsoft Purview data-loss and insider-risk signals.
- Write KQL queries to hunt across sign-in, device, and data-movement logs; automate evidence collection and case triage with PowerShell and Power Automate, eliminating repetitive manual steps.
- Use Microsoft Entra ID and Intune to confirm identity, access, and device posture; write up findings with a clear chain of evidence for HR, legal, and security leadership.

**Control Risks @ Meta** Oct 2024 – Aug 2026  
*Data Analyst — Investigation Enablement, Global Security & Investigations* Remote

- Engineered end-to-end ML pipelines in Python and SQL to track and recover over **\$200M** in lost and stolen Meta devices, enabling real-time asset recovery and reducing financial exposure at enterprise scale.
- Conducted forensic SQL investigations into organized retail theft incidents, identifying repeat offenders across multiple break-in events by cross-referencing behavioral patterns and transaction records in complex Meta databases.
- Designed and deployed LLM-powered dashboards to surface trends and generate actionable intelligence, reducing manual analyst review time and improving investigative throughput.
- Automated data collection and end-to-end investigative workflows using custom Python scripts, eliminating repetitive manual processes and accelerating case resolution time.
- Collaborated with cross-functional teams including legal, security operations, and engineering to implement scalable, data-driven solutions supporting global investigations.
- Presented findings and intelligence reports to senior stakeholders, driving informed decision-making on asset protection strategy.

## PROJECTS

**Online Music Database** | *SQL, Java, React.js*

- Designed and implemented a scalable RDBMS with optimized backend architecture supporting music playback, playlist management, and user interaction; applied advanced indexing and query optimization techniques to maximize performance under load.

**Project Helios** | *Python, Matplotlib, Data Visualization*

- Led a data visualization and analysis project using Python and Matplotlib to process solar flare datasets, uncover temporal trends, and deliver interactive visual reports with actionable insights.

## EDUCATION

**University of Houston** Class of 2024  
*Bachelor of Science in Computer Science — Houston, Texas*

## ACHIEVEMENTS

🏆 CodeRed UH Hackathon Winner — University of Houston 2025  
CodeCoogs Team Lead — Led student engineering organization 2024  
CodeCoogs Member — Active contributor to CS community 2022 – 2024

## VOLUNTEER EXPERIENCE

**Computer Science Association — University of Houston** Aug 2025 – Present

- Director of Marketing & Mentor — led HackHCC, HCC's first-ever hackathon; organize workshops and mentor CS students.